

Total No. of Questions : 8]

SEAT No. :

PD4104

[Total No. of Pages : 4

[6402]-64

S.E. (Artificial Intelligence and Machine Learning Engineering)

DISCRETE MATHEMATICS

(2019 Pattern) (Semester - III) (218541)

Time : 2½ Hours]

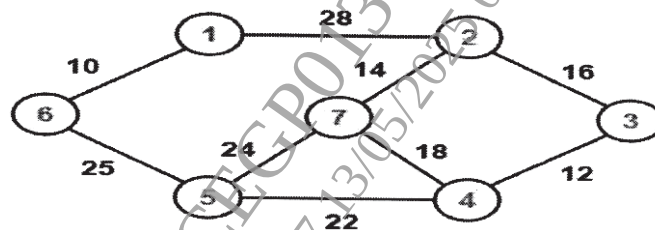
[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What is a planar graph? Suppose that a connected planar graph has six vertices, each of degree four. Into how many regions is the plane divided by a planar representation of this graph? [6]

b) Build a minimum spanning tree for the following graph using Kruskal's algorithm. [6]

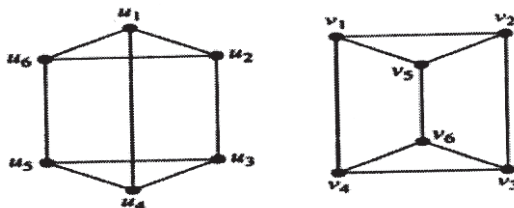


c) Define Prefix Code. Which of the following codes are prefix codes? Justify your answer. [6]

- i) a: 101, e: 11, t: 001, s: 011, n: 010
- ii) a: 010, e: 11, t: 011, s: 1011, n: 1001, i: 10101

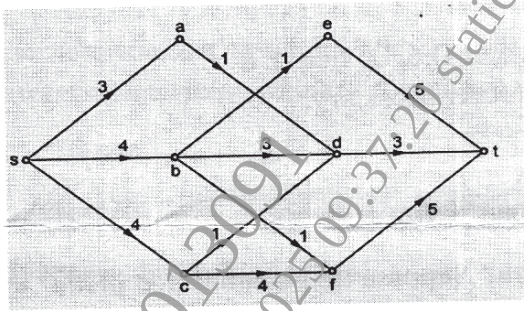
OR

Q2) a) Determine whether the following graphs are isomorphic to each other. Justify your answer. [6]

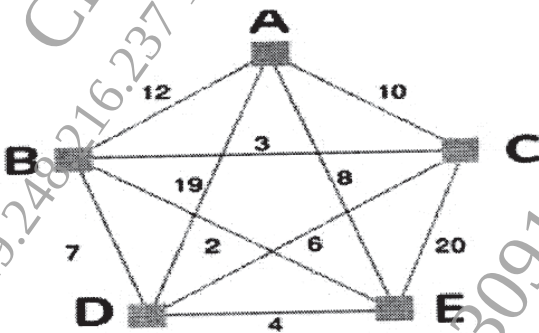


P.T.O.

- b) Using the labeling procedure, find the maximum flow in the following transport network. [6]



- c) Use the nearest Neighborhood method to solve the Traveling Salesperson problem starting with vertex A. Find the cost of the tour. [6]



- Q3) a) Consider these relations on the set of integers: [6]

$$R1 = \{(a, b) \mid a \leq b\}$$

$$R2 = \{(a, b) \mid a > b\}$$

$$R3 = \{(a, b) \mid a = b\}$$

$$R4 = \{(a, b) \mid a = 1 + b\}$$

$$R5 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R6 = \{(a, b) \mid a + b \leq 3\}$$

Which of these relations contain each of the pairs (1, 1), (1, 2), (2, 1), (1, -1), and (2, 2)?

- b) Solve the following recurrence relation. [6]

$$a_n = a_{n-1} + 2a_{n-2} \text{ where } a_0 = 2 \text{ and } a_1 = 7$$

- c) What is the minimum number of students required in a discrete mathematics class to be sure that at least five will receive the same grade, if there are five possible grades, A, B, C, D, and F? [5]

OR

Q4) a) Explain the injective function with an example? Determine whether each of these functions is an injection from $\mathbb{R}$ to $\mathbb{R}$. [6]

i) $f(x) = 2x + 1$

ii) $f(x) = x^2 + 1$

iii) $f(x) = x^3$

iv) $f(x) = (x^2 + 1)/(x^2 + 2)$

b) What is POSET? Let A is set of factors of positive integer m and relation is divisibility on A , i.e. $R = \{ (x,y) \mid x,y \in A, x \text{ divides } y \}$ For $m = 50$. Draw Hasse Diagram. [6]

c) Find the transitive closure by using Warshall's algorithm for the given relation as: [5]

$$R = \{(2,1), (2,3), (3,1), (3,4), (4,1), (4,3)\}$$

Q5) a) Using Binary expansion method solve the following (Show step-wise answer) $7^{71} \bmod 15$ [6]

b) Which of the following is true? Justify your answer. [6]

i) $8347 \equiv 597 \pmod{25}$

ii) $1960 \equiv 971 \pmod{23}$

iii) $1253 \equiv 1045 \pmod{10}$

c) Using Chinese Remainder Theorem, find the value of P using following data [6]

$$P \equiv 4 \pmod{11}$$

$$P \equiv 5 \pmod{7}$$

OR

Q6) a) What is a Mersenne prime number? Which of the following is the Mersenne Prime number? 7, 31, 255, 63, 8191 [6]

b) Find the Euler's totient function of the following numbers. [6]

i) 37

ii) 50

iii) 96

c) Find a multiplicative inverse of 52 mod 127 using Extended Euclidean Algorithm. [6]

Q7) a) Consider the set $A = \{1, 3, 5, 7, 9, \dots\}$ i.e. a set of odd positive integers. Determine whether A is closed under: [6]

- i) $a * b = a + 3b$
- ii) $a * b = a/b$
- iii) $a * b = a \cdot b$ (Multiplication)
- iv) $a * b = \text{power}(a, b) - b$
- v) $a * b = 2a + 2b$
- vi) $a * b = \min(1, a, b) - a$

b) Show that $(\mathbb{Z}_7, +)$ is an Abelian Group [6]

c) Prove that Hamming Distance $d(x, y) = d(y, x)$. [5]

OR

Q8) a) Show that with operation $*$ defined on the set $P = \{1, -1\}$ by $(a * b = a \cdot b)$ is an Abelian group [6]

$*$	1	-1
1	1	-1
-1	-1	1

b) Explain Integral Domain with an example. [6]

c) Consider the (2,6) encoding function e . $e(00) = 110000$, $e(10) = 101000$, $e(01) = 011110$, $e(11) = 111001$ [5]

Find the minimum distance of e .

